

An Efficient Deep Learning Enabled Multimodal Sentiment Analysis based on Neural Networks and Text Mining Architectures for Short-Form Social Media Data: A Comprehensive Analysis

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ABSTRACT

In recent years, massive amounts of short-form video content have been produced on TikTok, Instagram Reels, and YouTube Shorts, among others. This volume of content has changed the digital communications environment once again, this time emphasizing the utter dominance of communications in high-speed, Multimodal, and moving away from the former static and textual presentations. As a result, sentiment analysis, which previously dealt with a single textual and unimodal analysis has, evolved into a system that predicts the intended feeling from the integration of textual, acoustic and visual relationships, termed Multimodal Sentiment Analysis (MSA). Driven by breakthroughs brought about by deep learning applied to MSA, especially with the paradigm shifts introduced by Multimodal Transformers and

Cross-Modal mechanisms, among others, the state-of-the-art accuracy record has significantly improved. Advanced as these multimodal learning transformation technologies and architectures are, they incur significant computational costs, especially with the efficiency gap, self-attention, and large parameter counts. The increased costs and computational resources needed have always hindered wide implementation in real-time constrained edge environments. This research addresses two critical challenges simultaneously, and this survey examines these issues with its holistic review covering the period of 2016-Present with the MSA methods. A renewed focus on the articulation of meaningful efficiency in the trade-off in computational costs and system efficiency in the costs of computing systems is a noteworthy focus of this survey. The research proposes a definitive and rigorous set of taxonomies and a range of system architectures, termed efficiency-aware systems. These include, but are not limited to knowledge distillation, model pruning, and low-rank factorization that have been proposed to circumvent the latency challenges of mobile devices. Additionally, we assess both conventional scholarly repositories and newly arrived short-form video collections and highlight the instance distinction between laboratory results and actual field application (in-the-wild). The paper also identifies persistent challenges such as the “missing modality” issue and cross-lingual generalization while elucidating the future research directions with lightweight and robust MSA systems that can harvest high-density social media streams.

Keywords: Multimodal Sentiment Analysis, Deep Learning, Efficiency, Short-Form Video, Knowledge Distillation, Edge AI

Introduction:

In the past decade, the digital world transformed from predominantly text-based content to a multimodal, visually-oriented ecosystem. In the early stages of social media evolution, social media platforms (e.g., Facebook (now Meta) and Twitter) emphasized text-based microblogging, where users’ sentiments were articulated through individual texts [1]. As mobile wireless broadband access spread, a new genre of social media platform, TikTok, and its short video competitors, Instagram Reels and YouTube Shorts, gained extreme popularity. By 2025, social media videos, typically ranging from 15 to 60 seconds, have become the dominant form of massive global digital communication. The advancement in video media requires an adaptation of AI systems to analyze and integrate the complex interaction of linguistic content, acoustic prosody, and visual facial expression in real-time communication [2, 3]. In digital multimodal communication, the predominance of short videos poses problems to AI systems that rely on unimodal analysis, a system based on one communication medium (e.g., text) integrating several modules (e.g., semantic, syntactic, or pragmatic) in a pre-determined hierarchy. The text-only model fails to articulate complex emotional states (e.g., a digital discourse that is ironically named) in real time [4, 5]. The inconsistency between visual and audio modalities becomes a challenge that is heightened in shorts, given their content that becomes information dense, comes with vertical aspect ratios, rapid editing cuts, and can include other visual distractions, like text overlays, augmented reality filters, and background music [6, 7].

$$f_t = \mu + \rho GEGR_{it-1} + \vartheta^+ EU_{it-1}^+ \quad \text{Eq (1)}$$

These deficits exemplify why sentiment analysis is shifting from unimodal text-centered approaches to incorporating audio and visual modalities to address the issue. In a “reaction video,” for instance, the visual channel can provide additional information that acts as a corrective signal to the audio speech channel that allows for the contextual evidence to reframe the interpretation of what the speaker is literally saying [8, 9]. To be able to combine different types of information is very important if posts and footage from any social video platform are to have the sentiment of the post understood [10].

$$C_t = \vartheta^- EU_{it-1}^- + \phi^+ FD_{it-1}^+ + \vartheta^+ EU_{it-1}^+ \quad \text{Eq (2)}$$

The Accuracy-Efficiency Gap

Though theoretically essential, practical work in amateur and professional fields of multimodality is limited by a tension we term the "Accuracy-Efficiency Gap". In the academic domain of MSA, the main aim has been focused on maximizing predictive performance, which has been achieved through soaring deep learning models and architectural sizes [11]. The Transformer and its multimodal successors, namely the Multimodal Transformer and the Vision-and-Language Transformer, have achieved benchmark accuracies in the control of a myriad of tasks for the first time [12, 13].

The self-attention mechanism that is core to these architectures scales with quadratic complexity to the sequence length. A Multimodal Transformer with an architectural setup that includes 12 layers each with 768 hidden dimensions has over 150 million parameters which is notable and demanding a few gigabytes of available VRAM at inference time [14, 15].

These computational requirements represent a significant barrier to real-world applicability, especially for Edge AI. Consuming and producing social media almost exclusively occurs on resource-constrained mobile devices, not high-performance cloud servers [16]. Deploying a heavyweight Transformer model on a smartphone to conduct real-time sentiment filtering or recommendation generation may cause unacceptable latency, rapid battery drain, and thermal throttling. Thus, the central research question has shifted from questions of how to fuse modalities accurately to questions of how to fuse modalities accurately under hardware constraints [17, 18]. This urgency aside, the scholarly literature remains largely bifurcated: one stream focuses on heavyweight state-of-the-art models, while a smaller, fragmented stream attends to efficiency [19, 20].

$$I_t = \omega^+ TI_{it-1}^+ + \omega^- TI_{it-1}^- \quad \text{Eq (3)}$$

Scope and Focused Efficiency Analysis:

This survey aims to address these challenges with a more detailed examination of deep learning methodologies for performing multimodal sentiment analysis. Unlike other works that either survey sentiment analysis broadly or focus narrowly on the aspects of fusion, this survey incorporates the efficiency of the various frameworks

employed [21, 22]. A study of peer-reviewed articles from 2016-2025 sought to determine how the architectural choices impact the comprehension of the meaning and the speed of the system in drawing inferences. This survey attempts the first comprehensive analysis on the architectural design of MSA in terms of efficiency. This survey goes beyond the F1-score as the metric in computing model evaluations and assesses the model in terms of its FLOPs, the parameters the model contains, and the time taken to run the model.

Novel Taxonomy of Optimization and Short-Form Specificity: This survey proposes an integrated framework on "Efficiency-Aware" MSA, splitting the methods into Knowledge Distillation (Teacher-Student frameworks), Model Pruning, Quantization, and Factorized Fusion. This framework can be used to understand how to compress a model for deployment on devices with limited processing capability.

$$\Delta f_{t_i} = \sum_{j=1}^{n-1} \alpha_j \Delta GERG_{it-1} + \sum_{j=0}^{n-2} (\pi^{j+} \Delta EU_{it-j}^+ \quad \text{Eq (4)}$$

This analysis addresses the challenges associated with the social media content in short (TikTok format) videos with the disordered modalities in the lip-synced videos and other videos with missing modalities (auditory or visual). It also explores the extensive instances of multilingual switches within a conversation, especially in videos with the lip-synced modalities [23, 24].

Hardware-Aware Benchmarking: The work gives a balanced perspective as the performance data has been synthesized not only from high-end GPUs but also from mobile grade processors which benefits both academia and practitioners who seek to bring the MSA systems to the field [25].

$$\Delta f_{t_1} = \sum_{j=0}^{n-3} (\tau^{j+} \Delta FD_{it-j}^+ + \tau_j^- \Delta FD_{it-j}^-) \quad \text{Eq (5)}$$

In particular, the second section provides the necessary mathematics for this study, formally defines the multitask self-attention (MSA) phenomenon, and details the several pipelines for the targeted data to extract the textual, acoustic, and visual features. The third section highlights MSA research and its data collection, particularly in the noisy and short form data that is often encountered in real life, and surveys MSA comprehensively from short and open-access benchmarking datasets. The fourth section analyzes the critical analysis of state-of-the-art (SOTA) Deep Learning (DL) architectures that enhance efficiency and fusion paradigms. The fifth chapter summarizes explainable AI (xAI) and its challenges, especially the privacy-preserving Federated Learning, while the sixth chapter provides succinct recommendations for the development of next-generation eco-friendly (Green) MSA systems [26].

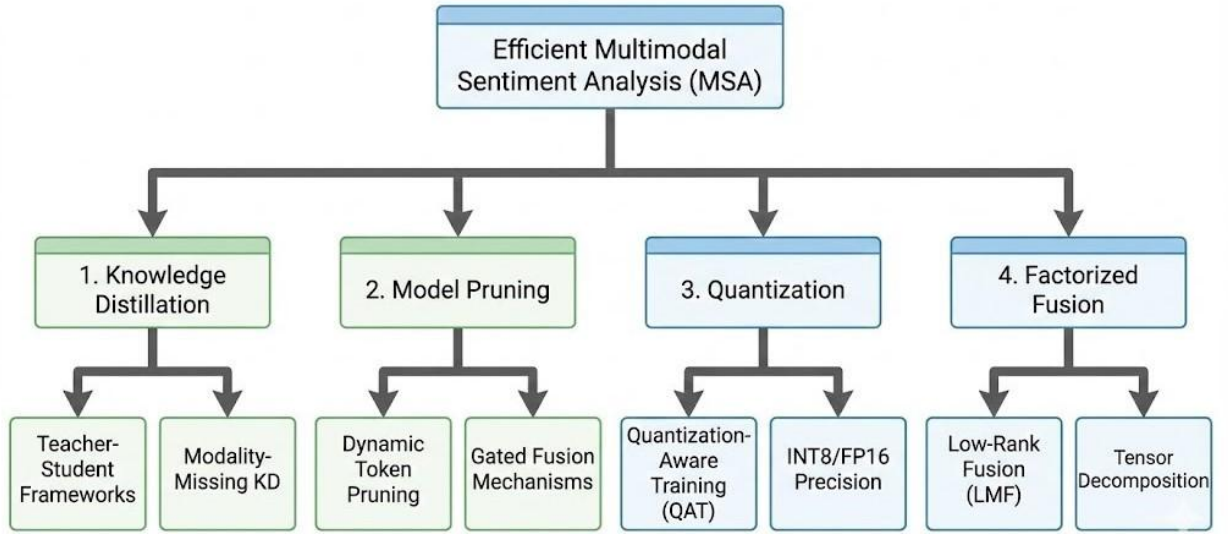


Figure 1: Taxonomy of Efficiency-Aware Multimodal Sentiment Analysis (MSA) [27].

$$\Delta f_{t_2} = \sum_{j=0}^{n_3} (\tau_j^+ \Delta F D_{it-j}^+ + \tau_j^- \Delta F D_{it-j}^-) \quad \text{Eq (6)}$$

Problem Formulation and Key Concepts

MSA differs from unimodal tasks in that the tasks dictate the joint processing of multiple heterogeneous data streams. These streams differ in their sampling rates and their dimensionality, as well as their information density and may be categorized as intermodal and intra-modal. We will order a discussion of the various fusion strategies in order from the most traditional to the most modern efficiency-oriented.

Mathematical Formulation

The MSA task can now be modeled as a formal supervised learning problem that targets sequentially connected, multimodal data. Let $D = \{(X^{(i)}, Y^{(i)})\}_{i=1}^N$ denote a dataset of N labeled video segments. Thereby, for each sample i , the input $^{(i)}$ is of triple modality: Language (l), Acoustic (a), and Visual (v).

$$\Delta f_{t_2} = \sum_{j=0}^{n_4} (\sigma_j^+ \Delta T I + \sigma_j^- \Delta T I_{it-j}^-) + \varepsilon_{it} \quad \text{Eq (7)}$$

$$X^{(i)} = (X_l^{(i)}, X_a^{(i)}, X_v^{(i)}) \quad \text{Eq (8)}$$

We represent each modality as a temporal sequence of feature vectors. Concretely, let $X_m \in \mathbb{R}^{T_m \times d_m}$, where $m \in \{l, a, v\}$, T_m , the sequence length T_m may be different for different modalities due to different sampling rates, and d_m represents the feature dimension for that modality [28, 29].

The goal now is to train a function $\mathcal{F}_\theta$, parameterized by weights θ , which maps such

heterogeneous inputs into a joint latent representation space to predict the target sentiment label $\hat{y}^{(i)}$ [30].

$$\hat{y}^{(i)} = \mathcal{F}_{\theta}(X_l^{(i)}, X_a^{(i)}, X_v^{(i)}) \quad \text{Eq (9)}$$

The target $\hat{y}^{(i)}$ typically assumes one of two forms: a discrete class label $y \in \{\text{Positive, Neutral, Negative}\}$ for classification tasks or a continuous intensity score $y \in \mathbb{R}$, (e.g., in the range $[-3, +3]$) for regression tasks. In the context of efficient MSA, the optimization objective is constrained [31]. The goal is to minimize empirical risk (loss) $\mathcal{L}(y, \hat{y})$, but also computational cost $\mathcal{C}(\mathcal{F}_{\theta})$ defined with respect to Floating Point Operations (FLOPs), memory footprint, and inference latency τ . As such, the optimum theoretical model is the one that satisfies the condition of $\tau \leq \tau_{max}$, and that is the most accurate, with τ_{max} usually being a latency budget of $< 50ms$ in the case of real-time applications [32].

Feature Extraction and Encoders: A Computational Analysis

Translating raw data signals to dense representation vectors is the first stage of every pipeline in MSA, and it is part of feature extraction. The type of encoder chosen for each modality determines the baseline computational cost of the entire system [33].

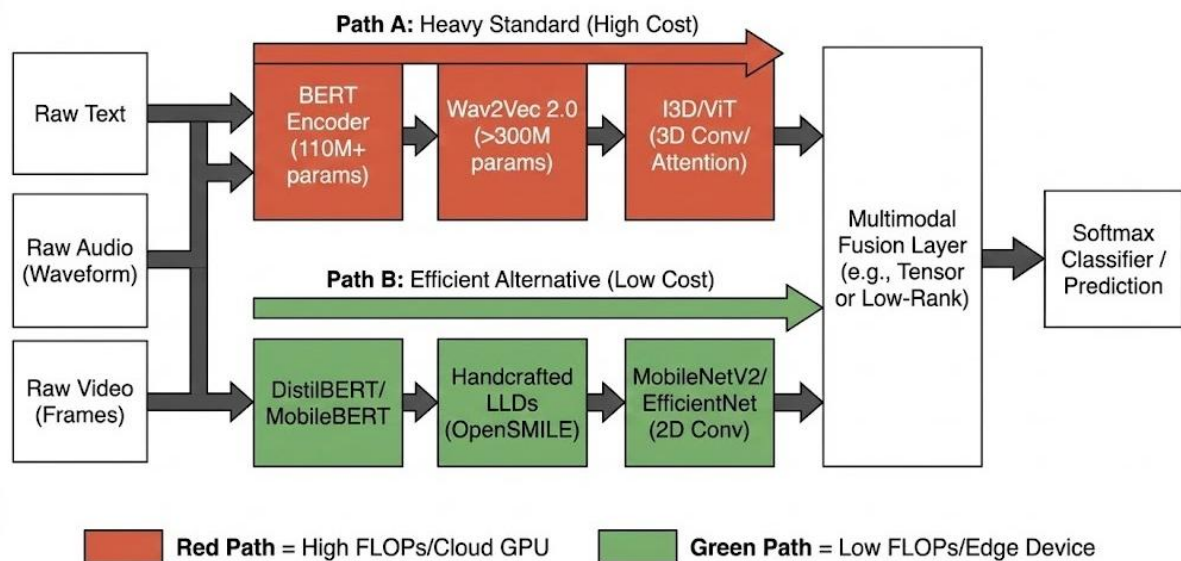


Figure 2: A comparative workflow of Standard (Heavy) vs. Efficient [34]

Linguistic Modality Encoders (X_l)

Textual features are derived from ASR transcripts or user-generated captions. Traditionally, static embeddings like GloVe have been utilized; these representations, however, capture no context-dependent meaning [35, 36]. In modern scholarship, PLMs, such as BERT, are considered the standard benchmark. BERT uses a multi-layer bidirectional Transformer encoder that allows dynamic word embeddings to capture highly complex syntactic and semantic relationships effectively [37, 38]. However, the BERT-base model has around 110 million parameters and therefore

requires massive computation resources, becoming an evident bottleneck for mobile deployment [39].

The Efficient Alternative and Acoustic Modality Encoders (X_a) : In order to balance these considerations, several distilled architectures have recently been proposed, including Distil BERT or ALBERT. BERT reduces the number of Transformer layers by half via knowledge distillation, preserving about 97% of BERT's performance, with a model that is 40% smaller and 60% faster [40, 41]. When it comes to short-form social media analysis-where text is usually informal and concise, for example, hashtags or slang-these lightweight encoders often provide a better trade-off between processing speed and semantic representation [42]. The auditory stream provides the prosody, pitch, and stylistic tone of the paralinguistic cues necessary to detect irony.

$$\delta_s = \left(\frac{m_x - m_n}{m_n} \right) \quad \text{Eq (10)}$$

$$\delta_v = m_x(\beta_r, \beta_g \beta_b,), \delta_{sv} = m_n(\beta_r, \beta_g \beta_b,) \quad \text{Eq (11)}$$

The Heavy Standard: Wav2Vec 2.0. Modern state-of-the-art acoustic modeling utilizes deep Transformer-based architectures such as Wav2Vec 2.0. These models are pre-trained on thousands of hours of unlabeled speech and learn to map raw waveforms directly to latent speech representations [43]. While highly accurate in noisy environments, they are computationally intensive normally comprising more than 300 million parameters-which limits their suitability for strictly resource-constrained settings [44].

The Efficient Alternative: Handcrafted LLDs. Efficiency-oriented methods make use of LLDs extracted using automated toolchains such as OpenSMILE. Such features consist of MFCCs, zero-crossing rates, and voice probability. The mathematical process involved in extracting the LLDs is light in terms of model depth with very low memory consumption. While they cannot provide the contextual depth that some of the Transformer-based approaches might, generally, they are sufficient for the purpose of recognizing high-arousal emotions like anger or excitement from short videos [45, 46].

Visual Modality Encoders (X_v) and The Heavy Standard: 3D-CNNs and ViT:

It follows that visual processing accounts for the most resource-intensive component in MSA due to high dimensionality of video data represented as (*Height* $\times$ *Width* $\times$ *Channels* $\times$ *Frames*) [47]. I3D (Inception-3D), and ViT, for example, use architectural mechanisms to capture spatiotemporal dynamics. These models represent video either as a volumetric block or as a sequence of tubelets, enabling them to model complex motion patterns. However, I3D uses computationally intensive 3D convolutions that come with very high FLOP counts, which quickly

deplete mobile power reserves [48, 49].

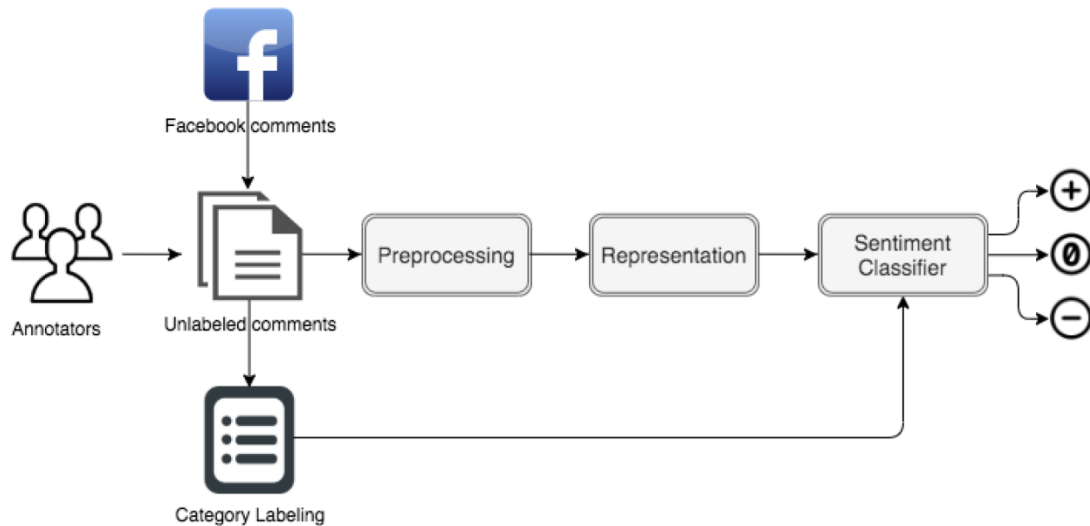


Figure 3: A Generic workflow for High-level architecture using 3D-CNNs and ViT [34]

$$R(t) = \sum_{i=1}^n FI_i(t) * \tau_{ih} + [\tau_h * r_{i-1}] \quad \text{Eq (12)}$$

The Efficient Alternative: 2D-CNNs and Mobile Net. Efficient approaches typically consider video as a sequence of static frames, where independent processing of frames is done using lightweight 2D-CNNs such as MobileNetV2 or EfficientNet. Mobile Net uses depth wise separable convolutions to drastically decrease the number of multiplications [50]. These methods extract features from key frames-such as taking one frame per second rather than features from the whole stream to obtain order-of magnitude reductions in computational load but with a possible risk of not detecting brief micro-expressions [51].

$$Y(t) = \omega[\tau_{ho} * h(t)] \quad \text{Eq (13)}$$

Taxonomy of Multimodal Fusion Strategies

Once features are extracted, the key challenge becomes that of Multimodal Fusion: integrating X_l , X_a , X_v into one representative vector. The adopted fusion strategy determines the model capability in capturing inter-modal dynamics-for example, the interaction between a visual smile and a sarcastic vocal tone [52, 53]. Early fusion, or feature-level fusion, implies the direct integration of unimodal features before classification. Mathematically, this is most often performed by concatenation: $Z = [X_l \oplus X_a \oplus X_v]$. The resulting combined vector Z is then fed into either a sequence modeling architecture, such as an LSTM or Transformer, or into a fully connected

layer [54]. A main advantage of early fusion is its simplicity and, it allows the model to learn low-level cross-modal correlations from the very beginning. However, it suffers from the "curse of dimensionality": the concatenation of high-dimensional vectors results in a big input space [55]. This increases the risk of overfitting and, usually, the computational cost of further layers. In addition, early fusion is often subject to modality dominance, where the model relies too much on the strongest signal, which happens to be textual most of the time, and ignores subtle cues coming from the audio or video [56].

Late Fusion (Decision-Level)

Late fusion, also known as decision-level fusion, works by training independent classifiers for each modality. Let $(y|X_m)$ denote the posterior probability distribution produced by the classifier corresponding to modality m . The final prediction is obtained through an aggregation mechanism, such as weighted averaging or majority voting: $\hat{y} = \operatorname{argmax} \sum w_m (y|X_m) \in \{l, a, v\}$

Late fusion allows for high efficiency and modularity; if one modality is not available-for example, a video without audio-the system can still function by using the other classifiers [57]. This quality especially makes it adaptable for 'in-the-wild' social media data. The fact that late fusion simply aggregates the final outputs of the different modalities ignores such rich feature-level interactions. This loss of interactions becomes especially problematic for capturing more sophisticated sentiments such as sarcasm, where the relationship of the signals is more important than their individual contributions [58].

Tensor-Based Fusion: The Complexity Bottleneck

To model explicit inter-modal interactions without the information loss of late fusion, [59] propose the Tensor Fusion Network (TFN). TFN calculates a Cartesian product of the unimodal feature vectors-a three-dimensional tensor Z : $Z = [X_l, 1] \oplus [X_a, 1] \oplus [X_v, 1]$. This model explicitly captures unimodal, bimodal, and trimodal interactions. However, the dimensionality of tensor Z is $d_l \times d_a \times d_v$. Since each dimension of the feature vectors is 100, the resulting tensor consists of 1,000,000 elements [60]. This exponential increase in dimensionality causes an exponential increase in parameters for the succeeding output layer, making TFN infeasible on edge devices because it represents ($O(d^3)$ complexity).

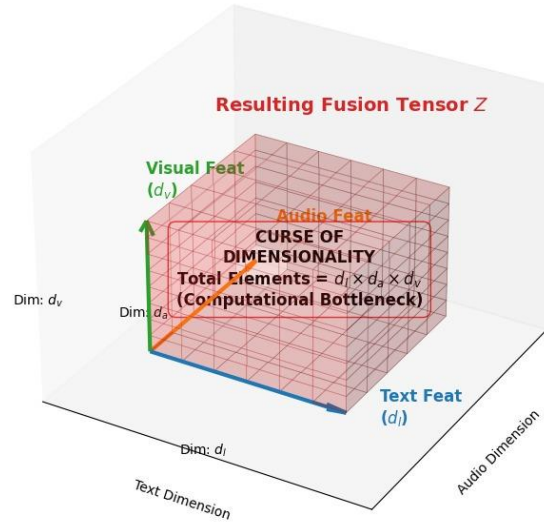


Figure 4: Visualization of Tensor Fusion & Dimensionality Explosion [61]

Low-Rank Fusion: An Efficient Approach

To mitigate the high cost of tensor fusion presented a Low-rank Multimodal Fusion (LMF) framework. LMF utilizes low-rank tensor decomposition and specifically CANDECOMP/PARAFAC (CP) decomposition to approximate the huge weight tensor of Tensor Fusion Network (TFN), by a set of low-rank factors [62]. Rather than explicitly computing the high-dimensional tensor Z , LMF computes the output via parallel factor matrices, which decreases the computational complexity from exponential to linear in d , ($O(d)$). Decoupling fusion weights allows LMF to reach a comparable benchmark to TFN with a reduction of weights by over 90%, making LMF a benchmark cycle for compact memory in multimodal learning [63].

$$\omega = E_f * \frac{1}{1 + e^{-\theta t_f}}$$

Eq (14)

Datasets and Benchmarks

To accurately evaluate MSA architectures, one must have access to well-documented and high-quality datasets. These datasets should reflect the richness and intricacies of human interaction [64]. While academic foundational datasets have traditionally driven the area, the unique characteristics of short-form social media-narrow aspect ratios, high levels of noise, and informal language-mean that specialized collections are needed to evaluate performance in realistic scenarios. This section surveys standard benchmarks and emerging social-media-specific datasets, before outlining the efficiency metrics required for edge-deployable model evaluation [65].

These datasets form the main testbeds for the evaluation of Deep Learning architectures in terms of their accuracy. They are defined by high production values

and clear audiovisual signals, offering a controlled environment for algorithm development [66].

CMU-MOSI (CMU Multimodal Opinion Sentiment Intensity)

The CMU-MOSI dataset of [67] is the de facto standard benchmark for MSA (see Table 1 for specifications). While seminal in its time, MOSI has relatively few examples and highly controlled audio-visual conditions—a centered face and little to no background noise—that do not capture the chaotic variability of modern short-form content. As such, MOSI remains the standard for evaluating the baseline capability of a fusion mechanism to align linguistic and visual cues in ideal settings.

CMU-MOSEI (CMU Multimodal Opinion Sentiment and Emotion Intensity)

As the follow-up to MOSI, CMU-MOSEI [68] is the largest available dataset of its kind in the research community. Compared with MOSI, most videos are horizontal (16:9) and were recorded either in professional or semiprofessional settings, which are quite different from typical user-generated content on mobile platforms.

MELD (Multimodal Emotion Lines and Short-Form Dataset)

The MELD dataset shifts the focus from monologues to dyadic and multi-party interactions. Extracted from the TV series *Friends*, MELD is quite different from the MOSI/MOSEI isolated clips; its samples are embedded in conversational contexts where the sentiment of a given utterance is largely determined by the previous turns. This dataset is thus crucial for evaluating Context-Aware models that have to track emotional variations over time. While it is highly relevant for the analysis of “duet” or “stitch” videos appearing on social media platforms, where users interact with each other in turn, it is based on acting rather than spontaneously produced natural behavior, which may limit its generalizability to real-world user behavior [69, 70]. Recent studies have therefore introduced datasets, curated especially from social media platforms, to bridge the gap between academic inquiry and practical deployment. Such datasets naturally come with their unique challenges, inherent in their in-the-wild nature for mobile content creation [71].

TikTok-24k and Short-Form Video Datasets and Vertical Aspect Ratio (9:16):

Other emerging datasets in this category, like TikTok-24K, focus exclusively on vertical, shortform videos, typically lasting between 15–60 seconds. These collections show some distinctive characteristics that challenge traditional modeling approaches: Unlike the horizontal frames of YouTube and television, vertical video tends to focus on a small area around the face or body. However, large parts of the frame are still regularly occluded by elements from the user interface—for instance, heart icons, overlays of comments, and text captions—which complicates the resistance of visual encoders against partial occlusion [72, 73].

Augmented Reality (AR) Noise and Acoustic Complexity: A large proportion of the short-format videos apply AR filters that either distort facial landmarks—large eyes, smooth skin, etc., or attach virtual stickers. The traditional face-based emotion recognition models are trained with clean data and fail to extract robust affective features in the presence of these altered faces [74, 75].

The audio signals of these datasets are also heavily noisy, and often have copyrighted music overlays masking speech, or lip-sync situations in which the audio does not correspond to the speaker. This setting calls for effective fusion strategies able to dynamically down-weight the acoustic modality when it is irrelevant or unreliable [76].

Table 1: Comparison of Standard vs. Social Media MSA Datasets

Dataset Name	Source Platform	Size / Samples	Modalities	Key Characteristics	Usage Type	Ref
CMU-MOSI	YouTube	2,199 segments	Text, Audio, Video	Monologue, controlled environment, centered face.	Academic Benchmark	[77]
MVSA	Twitter	4,000+ pairs	Text, Image	Static. Image-text pairs, checking for incongruity.	Static Analysis	[78]
CMU-MOSEI	YouTube	23,500 segments	Text, Audio, Video	Large scale, diverse topics, emotion labels included.	Academic Benchmark	[79]
MELD	TV (Friends)	13,708 utterances	Text, Audio, Video	Dyadic/Multi-party conversational context.	Context-Aware	[80]
Fakeddit	Reddit	>1 Million	Text, Image	High metadata (upvotes), multimodal misinformation.	Scalability Testing	[81]
TikTok-24k	TikTok	~24,000	Text, Audio, Video	Vertical (9:16), AR filters, heavy background noise.	Social Media / In-the-Wild	[82]

Social Media and Short-Form Specific Datasets

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$$f_{th} = \sum_{j=0}^h \frac{\delta_j}{\delta EU_{it-1}^+}, m_h^- \quad \text{Eq(15)}$$

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MVSA (Multi-View Sentiment Analysis) and Fakeddit and Multimodal Misinformation

The MVSA dataset is extracted from Twitter and represents static multimodal content, where image and text pairs are formed. It has two subsets, MVSA-Single and MVSA-Multiple, distinguished by the number of annotators per pair. Without the temporal aspect embedded in the video, MVSA nonetheless proved to be an important benchmark for assessing matching of image-text pairs and the task of detecting "text-image incongruity," where the text is neutral but the intended image reveals a strong negative feeling—a picture of a disaster paired with a factual caption [87, 88].

Fakeddit, although created primarily for fake news detection, is a large multimodal dataset originated from Reddit, increasingly used for testing the scalability of MSA systems. It consists of text, images, and rich metadata, such as comments and upvotes. This resource provides researchers with the opportunity to explore the incorporation of Social Context Features like user credibility scores and engagement metrics within the sentiment analysis pipeline [89, 90]. For efficient MSA, Fakeddit serves as a practical benchmark for handling very large-scale data streams where it is computationally infeasible to process every sample with a heavyweight Transformer model [91, 92].

Performance Metrics for Efficient MSA

Within the scope of this survey, assessing models solely by their prediction accuracy is insufficient. A proper review of the viability of MSA architectures for edge deployment calls for a broader set of efficiency metrics.

Table 2: Heavy Standards vs. Efficient Alternatives

Modality	Heavy Standard (SOTA)	Drawback (Computational Cost)	Efficient Alternative	Benefit
Linguistic	BERT-base [93]	~110M params; massive memory.	DistilBERT [93] / Mobile BERT [94]	40% smaller; 60% faster inference.
Acoustic	Wav2Vec 2.0 [95]	>300M params; slow on edge.	Handcrafted LLDs (OpenSMILE [96])	Low memory; sufficient for arousal.
Visual	I3D [97] / ViT [98]	High FLOPs (3D convs/attention).	MobileNetV2 [99]	Depth-wise separable convs; fewer ops.
Fusion	Tensor Fusion (TFN) [100]	Exponential complexity (d^3).	Low-Rank Fusion (LMF) [101]	Linear complexity (d); weights decoupled.

Inference Latency (τ)

Inference latency measures the time required to process one video sample from input to prediction, often expressed in milliseconds (ms). For real-time applications, such as content moderation during a live stream, for example - it is generally considered that $\tau < 100$ ms serves as the upper bound for acceptable performance [101]. Latency depends on the underlying hardware platform, and strong benchmarking should strive to report results on representative target hardware, such as ARM-based mobile processors, rather than high-end server GPUs exclusively, like the NVIDIA A100 [102].

FLOPs (Floating Point Operations) and Model Size and Parameter Count

FLOPs are a hardware-agnostic measure of a model's computational complexity. They indicate the total number of floating-point multiplication and addition operations needed to make one forward pass through the network. Lower FLOP counts mean lower power usage, which is essential for mobile devices running on batteries. Most efficient MSA models tend to cut FLOPs down from the gigaflop range of billions, as seen in Transformers, to the megaflop range of millions [103, 104]. Model size is defined by the number of trainable parameters (weights) within the network, conventionally expressed in millions (M), and the associated storage footprint in megabytes (MB). This metric defines whether a model can reside in RAM or on-chip cache of an edge device. As an example, a standard BERT-base model (~440 MB) can easily exceed the available memory of a mid-range smartphone when sharing with other active applications. That is what encourages compression methods such as quantization, as they are able to minimize footprint size, for instance to less than 50 MB, with minimal loss in accuracy [105].

$$\text{minimum} f d_k(x) = \sum_{m=1}^{C * M} \frac{1}{n} f d_m(x)$$

Eq (16)

Deep Learning Architectures for Efficient Multimodal Sentiment Analysis

This section extensively reviews modern deep learning architectures that have been proposed for MSA. The literature is divided chronologically and structurally, from the analysis of static images and texts, then to the processing of dynamic videos, and finally to the emerging model of “Efficiency-Aware” architecture. The discussion integrates a computational review of the tradeoffs between parameters, inference delays, and memory assessments to determine the relevance of these frameworks in the application of short-form social media [106].

Static Multimodal Analysis - Image and Text and Co-Attention and Aspect-Level Fusion

The early works of MSA focused on static content, like tweets or Instagram posts, in which the input consists of a single image with an associated text caption. While these models are relatively less computationally expensive compared to video-based approaches, they created necessary mechanisms for feature alignment and cross-modal attention which modern video-based systems rely on [107]. Early methodologies utilized simple Early Fusion using concatenation; however, significant works started to appear with the introduction of Co-Attention networks. “Co-Attention Network, CAN”, initially developed for “Visual Question Answering, VQA”, and later adapted for sentiment analysis [108]. The authors propose a hierarchical co-attention layer where the image representation informs the textual attention and vice-versa, where textual cues highlight important visual aspects. By employing the model to generate a series of attention mechanisms in an alternating manner, irrelevant background noise of images is effectively suppressed—such as ignoring crowds to focus on a face—and particular visual regions are aligned with sentiment-bearing lexical items. In effect, this serves to reduce the effective feature dimensionality before fusion, which is an early form of efficiency [109, 110]. Introduce VistaNet, a visual aspect attention network. The authors argue that visual features are not just complementary but actually define the aspect of the sentiment. For instance, an image of a bruised fruit tells the model that the negative textual content is about the product quality, not about the price of shipping. VistaNet aligns visual regions with textual sentences using a soft attention mechanism and performs well on visual review datasets like Yelp. However, these models all rely on heavyweight backbone architectures; for example, ResNet-152, though it has very high accuracy, results in high latency during feature extraction [111, 112].

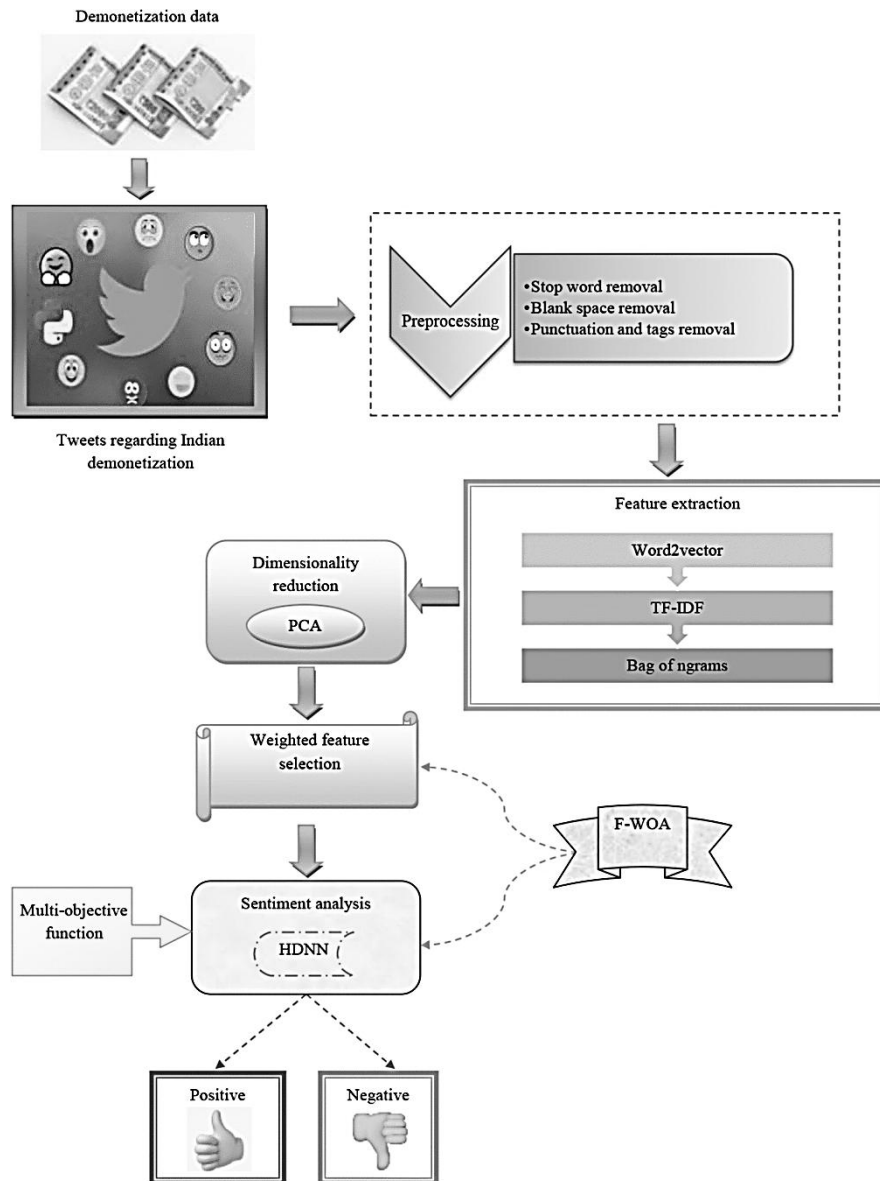


Figure 5: A Generic workflow based on Deep Learning Enabled Multimodal Sentiment Analysis [113]

Transformer-Based Static Fusion and Dynamic Multimodal Analysis:

With the appearance of BERT, static MSA shifted toward Transformer-based architectures. The authors in [114] introduce a dual-stream framework coupling a pre-trained ResNet-50, for image feature extraction, with a BERT encoder for text processing. The main contribution lies in an ABF layer, which performs dynamic weighting of the modalities. By learning, when textual input is brief, or ambiguous, as in most social media posts being mere hashtags, the model gives higher weight to the visual stream. While effective, employing two very large pre-trained models running in parallel (ResNet and BERT) presents a serious memory bottleneck, often surpassing 500 MB and challenging the storage of mobile applications [115].

Then came the transition to short-form video with its intrinsically temporal dimension, calling for architectures capable of modeling sequential dependencies across three heterogeneous streams. This domain so far has been shaped by two successive waves of innovation-the Tensor Fusion paradigm and the Transformer paradigm [116]. The dominant paradigm before 2019 relied on using an LSTM network to model the temporal sequences with complex fusion layers on top of them. The notable work done by Zadeh et al. (2017) proposed the Tensor Fusion Network, or TFN for short, which was designed to model intermodal interactions explicitly. This would represent $(Z = h_l \otimes h_a \otimes h_v)$, through the outer product of the feature vectors from text, audio, and video. Mathematically speaking, this operation captures unimodal, bimodal, and trimodal dynamics without needing to learn weights on every single interaction [117, 118].

$$g^t(x) = \sum_{i=1}^t y_i \prod_{j=1, j \neq i}^t \frac{(x - x_j)}{(x_i - x_j)}$$

Eq (17)

Critique of Efficiency: While TFN does a great job of capturing interactions, it suffers from "Curse of Dimensionality." The tensor dimension becomes the product of the input dimensions $(d_l \times d_a \times d_v)$. In case of modality feature sizes 128, the fusion tensor has more than two million elements. Due to this exponential growth, the number of parameters in further fully connected layers will be huge, which makes TFN not suitable for mobile devices with limited RAM [119]. addressed this scalability challenge by proposing the Low-rank Multimodal Fusion (LMF) framework. LMF performs low-rank tensor factorization-specially, CANDECOMP/PARAFAC decomposition-to decompose TFN's large weight tensor into modality-specific low-rank factors, hence reducing the computational complexity from exponential to linear-that is, $(O(d))$. Empirical results on CMU-MOSI show that LMF achieves accuracy comparable to TFN while reducing the number of parameters by more than 90%, making it one of the first truly efficient fusion mechanisms in the literature [120, 121].

The Transformer and Cross-Modal Attention Epoch

The modern SOTA is defined by Multimodal Transformers that avoid RNNs and instead rely on the global attention mechanism. [122] introduce the concept of the Multimodal Transformer, called MulT. The architecture introduces the concept of Cross-Modal Attention, in which the Query (Q) of one modality attends to the Key (K) and Value (V) of another modality for example, the Text attending to Video. This element implicitly solves the alignment problem implicit in asynchronous data-for example, finding a spoken word within its corresponding visual gesture, which may come several milliseconds later-without the need for computationally expensive explicit preprocessing [123]. MulT utilizes multiple layers of pairwise, directional cross modal attention (Text $\rightarrow$ Audio, Audio $\rightarrow$ text), and so on-and captures deep

semantic alignment [124].

Efficiency-Aware Architectures

Critique of Efficiency: While accurate, MulT is also very resource-intensive. The self-attention mechanism has quadratic computational complexity as a function of sequence length ($O(L^2)$) [125]. Furthermore, typically, MulT utilizes up to six cross-modal transformer blocks along with three self-attention blocks, resulting in parameters exceeding 160 million often. For the typical 60second TikTok video, standard CPU-based inference time could go over 500 ms, affecting real time scrolling feed feasibility. In [126], the authors introduce ViLT, a Vision-and-Language Transformer, which aims to simplify the pipeline by directly feeding raw image patches into a Transformer and thus bypassing the expensive CNN-based feature extraction (e.g., ResNet). While this speeds up the feature extraction stage, the backbone of the Transformer remains large. Since it relies on heavy self-attention layers, although the models achieve state-of-the-art results, like F1-scores over 85% on MOSEI, they actually fail the "Edge AI" criterion in that cloud-grade GPUs are a must for reasonable latency [127]. As summarized in Table 3, we categorize these optimization strategies into four distinct pillars: Distillation, Pruning, Quantization, and Factorization.

Table 3: Taxonomy of Efficiency-Aware Optimization Techniques

Category	Technique	Core Mechanism	Primary Benefit	Ref
Distillation	Teacher-Student	Train a small model to mimic a pre-trained heavy "Teacher."	Reduces model size while retaining accuracy.	[128]
Pruning	Dynamic Token Pruning	Identify and skip processing for redundant video tokens (e.g., background).	Reduces FLOPs by skipping useless computations.	[129]
Quantization	QAT (Quantization Aware Training)	Reduce weights from FP32 (32-bit) to INT8 (8-bit).	4x reduction in memory footprint.	[130]
Architecture	Factorization	Decomposing high-rank tensors into low-rank factors.	Reduces parameter count significantly.	[131]

Knowledge Distillation Teacher–Student Frameworks

KD typically involves training a small "Student" model to mimic the behavior of a large, pretrained "Teacher." [132] introduce Mobile BERT as a compressed version of BERT for resource-constrained devices. In the context of MSA, Mobile BERT acts like a lightweight textual encoder. The authors implement a progressive knowledge distillation strategy wherein the knowledge is transferred layer by layer from a deep teacher to a slim student. It achieves a model size reduction of 4.3× and inference

acceleration of $5.5\times$ compared to BERT-base with negligible accuracy loss on sentiment tasks. More recently, [133] proposed a Modality-Missing Knowledge Distillation framework. This approach mitigates the high processing cost of video frames by training a heavy multimodal teacher (Text + Video + Audio) to supervise a lightweight student model, which takes only text during inference. This student learns to "hallucinate" the visual features it would expect for the given textual input. The eventual deployed system then executes at unimodal speeds-processing only text, which is very fast-while retaining the robustness of a multimodal system. It finds especially good performance on short-form videos, where often the caption summarizes the visual content [134, 135].

Pruning and Dynamic Computation

Pruning methods seek either to eliminate superfluous parameters or avoid redundant computation dynamically. [136] propose a Gated Fusion mechanism for dynamic modality selection. The model uses lightweight policy networks that evaluate the information content of each video frame in real time. If a frame is considered redundant-for example, a static background or black screen-or when the accompanying audio is silent, the gate turns off the computation for the respective branch. Experiments performed on the TikTok-24k dataset show FLOPs reduced by 40% due to the skipping of processing for un-informative segments. [137] present Dynamic Token Pruning for Transformers. In a video transformer, not all spatial patches carry useful information; for instance, the corners of the frame often represent background. This approach detects and abandons useless tokens in the early layers, saving them from consuming compute cycles in the deeper layers. The method achieves a $2x - 3x$ acceleration in the processing of videos with less than a 1% drop in accuracy [138].

Quantization-Aware Training (QAT)

Quantization decreases the model weights' precision from 32-bit floating-point (FP32) to 8-bit integers (INT8). [139] proposed an all-encompassing framework for Quantization Aware Training (QAT) applied to Transformers. Unlike PTQ, which may result in performance degradation, QAT simulates quantization noise during training, hence allowing the model to adapt to lower precision. When applied to multisensory attention (MSA), researchers have successfully quantized both the feature extractors-e.g., Mobile Net-and the fusion layers [140]-e.g., Transformer. This compression reduces the model size by approximately $4\times$ (e.g., from 400 MB down to 100 MB) and allows exploiting specialized integer arithmetic units-such as DSPs or NPUs available on modern smartphones-doubling inference throughput without changing the model architecture [141].

Factorized and Lightweight Architectures

Some works forego compression of heavy models and instead pursue the design of efficient architectures from scratch. In [142] TinyM2Net is proposed as a flexible CNN-based architecture for on-device multimodal sentiment analysis. The authors substitute standard convolutional blocks with Depth-Wise Separable Convolutions popularized by Mobile Net that factorize a conventional convolution into a depth-wise

and a point-wise layer, reducing computation by a factor of 8 to 9. Furthermore, they introduce a “Factorized Bilinear Pooling” layer that approximates high-order fusion using low rank matrices [143]. The system achieves real-time inference (< 20 ms) on ARM-based mobile processors, thus validating its suitability for live social media mining. In [144] the authors introduce the Light-Multimodal model, which is based purely on the MLP architecture of mixers, and neither uses Convolution nor Attention. Hardware-friendly, this model uses only simple matrix multiplications, highly efficient, although at the cost of some capacity in modeling long-range temporal dependencies compared with Transformer-based approaches [145].

Comparative Analysis of Architectures

The synthesis of the current landscape shows that a trade-off frontier exists. Multimodal transformers, such as MulT and ViT, top academic benchmarks yet remain impractical due to their complexity for edge deployment [146-150].

Medium-Accuracy / Medium-Efficiency: Tensor-based approaches, such as LMF, and Hybrid RNNs represent an intermediate solution but still struggle with high-resolution video inputs. Distilled (Mobile BERT), Quantized, and Factorized (TinyM2Net) models represent the potential future direction of short-form MSA. While they may lag the SOTA in F1-score by about 1–2%, their ability to operate in real time (latency < 50 ms) makes them the only deployable candidates on social media platforms [151-153].

Table 4: Summary of Key Literature (2016-2025)

Ref	Proposed Model	Focus Area	Key Contribution
[154]	TFN (Tensor Fusion)	Accuracy (Interaction)	Explicit modeling of trimodal interactions via outer product.
[155]	LMF (Low-Rank Fusion)	Efficiency (Fusion)	Decomposed large tensors to achieve linear complexity.
[156]	MulT (Multimodal Transformer)	Accuracy (Alignment)	Cross-modal attention to handle unaligned data.
[157]	MobileBERT	Efficiency (Text)	Compressed BERT for mobile devices via distillation.
[158]	Gated Fusion	Efficiency (Video)	Dynamic skipping of redundant frames to save power.
[159]	Modality-Missing KD	Robustness/Efficiency	Student model hallucinates missing visual features from text.

Challenges and Future Directions

Regardless of the breakthroughs in Deep Learning architectures, as well as the development of optimally efficiency-focused improvements, MSA is faced with a number of problems preventing its large-scale application in the real world [160, 161]. The transition from controlled academic settings-e.g., CMU-MOSI-to the

dynamic high-velocity milieu of short-form social media presents robustness, interpretability, and generalization challenges that are yet to be resolved by current SOTA models. This section illustrates the open problems and presents rigorous avenues for future research, focusing on the limits of edge deployed systems [162, 163].

Robustness to Incomplete Data

In theoretical experimental settings, multimodal datasets are carefully constructed such that each data point contains full, aligned streams of text, audio, and video. However, in real-world applications like TikTok and Instagram Reels, Modality Dropout is everywhere. A large fraction of user-generated content is incomplete: a user might upload a slideshow of photos with background music but without speech-missing linguistic features, a video with mute or missing sound because of copyright issues- missing acoustic features, a text-only "story" over a black screen-missing visual features [164, 165]. Most state-of-the-art Multimodal Transformers (e.g., MulT, ViLT) assume complete data availability. It is common to observe catastrophic failures for those models, or even rely on the use of manually inserted zero vectors when a modality is missing, substantially harming cross-modal attention performance through the injection of noise into the fusion tensor. This inability to handle missing data will, in turn, often require maintaining separate fallback models at the edge-for example, a text-only model for mute videos, increasing storage at the edge [166, 169].

Future Directions: Generative Imputation and Modality-Invariant Training

Future research efforts should be channeled into Robust Representation Learning. The two promising avenues going forward.

Modality-Invariant Training and Generative Imputation:

Methods where models are randomly exposed to missing modalities during training, for example, stochastically masking the audio channel in 30% of training batches. This mechanism forces the network to learn robust features that do not get overly dependent on any single modality, effectively emulating dropout conditions similar to real-world deployment. The inclusion of lightweight generative models, such as Variational Autoencoders or Generative Adversarial Networks, that create or "hallucinate" missing features based on the available modalities. For example, given a visual stream that is present and audio that is absent, a cross-modal generator could infer the likely acoustic feature from facial expression; thus, keeping the integrity of the input tensor during inference and cohering the fusion pipeline.

Cross-Lingual Generalization and Code-Switching

Current MSA benchmarks are predominantly English-centric, reflecting a bias in academic resource creation. However, short-form video platforms are inherently global, and the linguistic landscape of user-generated content is highly heterogeneous. A specific challenge in this domain is Code-Switching—the practice of alternating between two or more languages within a single utterance or conversation (e.g., "This outfit is so kawaii, I absolutely love it!").

Standard Pre-trained Language Models (PLMs) like BERT are typically monolingual and struggle to parse the semantics of mixed-language text. When a monolingual

English model encounters a non-English token (e.g., kawaii), it often treats it as an "unknown" token (<UNK>), resulting in a loss of semantic information. Furthermore, deploying separate models for every language is computationally infeasible for mobile applications due to storage constraints. The integration of Multilingual Encoders (e.g., XLM-RoBERTa or mBERT) into efficient MSA pipelines is urgent. However, simply replacing BERT with mBERT increases model size. A more efficient direction lies in Adapter-based Tuning [139]. In this paradigm, the massive pre-trained multilingual model is frozen, and only small, language-specific bottleneck layers ("adapters") are inserted between the transformer layers. This allows a single core model to adapt to new languages (e.g., Spanish, Hindi, Hinglish) with minimal parameter overhead (often <5% of the total model size), preserving the efficiency required for mobile deployment while enabling robust handling of code-switched content. There is a growing tendency towards adopting models that have greater optimization and efficiency. These 'Student' models and architectural compression involving deep neural networks act as black boxes. In sensitive use cases – such as suicidal ideation and hate speech detection, or radicalization on social media – there is a need to justify, to stakeholders, why a given model assessed a video and decided it was negative. Was it due to a keyword used, a micro-expression in the face, or a tone that was sarcastic and acrimonious. The black box aspect of such efficient models is seen to be detrimental from a trust perspective and, at the same time, limits one's capacity to offer a solution to algorithmic fairness when, for instance, a model is seen to continuously rate a given dialect as 'hostile' and 'aggressive' to a large extent.

Attention Visualization and Counterfactual Analysis: Lightweight approaches should be developed to visualize attention Rollout of fusion layers and trace which video frames or text tokens contributed to the prediction. Visualizations can be rendered as "heat maps" to show which drivers of sentiment were impactful. A more advanced approach is to develop synthetic variations of a video sample (i.e., freeze the text and change the voice pitch to another voice [alter the pitch] and observe differences in the model prediction. It is possible to provide causal explanations of model behavior using these counterfactuals while remaining computationally efficient, as real-time inference does not involve additional costs in prediction. Involving the analysis of the user's biometric and emotional information, such as voice and face recordings, is problematic. The customary cloud-based strategy suggests the uploading of live video streams to the cloud, which raises privacy, surveillance, and bandwidth costs and latency issues. Whereas, having the processing exclusively on device protects privacy but stagnates improvement of the model as the users are unable to learn from the data as a whole. Federated learning is a decentralized method that maintains the trade-off between privacy and continual learning. It computes gradients locally based on the user's interaction data and sends only the weight updates (not the raw video data) to a central server. The server aggregates these updates (e.g., via Federated Averaging) to improve the global model, which is then sent back to the devices. Future research must address the challenge of System Heterogeneity in FL. A global model should have the ability to collect and synthesize updates and data from an array of devices and systems within an ecosystem without performance degradation caused by device disparity. Systems like

Split Learning, in which the uppermost and most computationally intensive model layers are kept on the server, while the remaining and comparatively less intense higher layers are kept on the client, could provide a midpoint for situations where the edge device is too resource-deficient to complete an entire backpropagation.

Conclusion and Recommendation

The shift from text-based microblogging to high-speed, short-form video presentation has established Multimodal Sentiment Analysis (MSA) as a base technology to grasp public opinion and consumer behavior, and to analyze social behavior. This survey has tackled the problem rigorously and systematically, and unlike previous works, has focused primarily on computational efficiency. Furthermore, the survey has provided a detailed taxonomy of Deep Learning of the Early Fusion and Tensor-based models to the current age of what is referred to as 'Efficiency-Aware' models. A clear "Accuracy-Efficiency Gap" emerges from the analysis. Although several Multimodal Transformers, such as MulT and ViLT achieve high accuracy levels, they do so because of computationally inefficient cross-modal attention ($O(L^2)$) (i.e. quite high); and as such, they are not suitable for real-time applications on the social media edge devices which are quite resource-poor. On the contrary to other reviews, this review has noted that the future of scalable MSA rests on the two fronts of optimization and compression. This would be via Knowledge Distillation, LowRank factorization, and Quantization-Aware Training which are vital techniques. The use of these strategies means that the implementation of "Green AI" models is possible, as they preserve considerable semantic understanding, while using only a fraction of the computational and energy resources. Over the next decade of research, the problem is building models that are of real-world use and demonstrate utility, particularly in the presence of noise and missing data, as well as the diverse linguistic ecosystem in which they must operate, rather than developing models of little more than incremental utility, particularly in terms of F1-score improvements on clean benchmarks of a task. With the open problems of cross-lingual generalization, missing modalities, and privacy preserving Federated Learning, the research community can develop robust MSA systems that provide emotional intelligence to the billions of edge devices in the network.

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